

COMP 2121
DISCRETE
MATHEMATICS

Quiz 5

Name:

Student ID:

Set: A B C D (circle one)

- Time allowed: 20 min

1. (1.5 marks) A given tree has:

- 6 vertices of degree 5,
- 12 vertices of degree 4,
- 9 vertices of degree 3,
- x vertices of degree 1.

$$\sum_{i=1}^{|V|} \deg(v_i) = 2 \cdot |E|$$

$$|V| = |E| + 1$$

Determine the value of x .

$$6(5) + 12(4) + 9(3) + x(1) = 2|E|$$

$$105 + x = 2|E|$$

$$105 + x = 2(26 + x)$$

$$105 + x = 52 + 2x$$

$$105 - 52 = 2x - x \Rightarrow \underline{\underline{x = 53}}$$

$$6 + 12 + 9 + x = |E| + 1$$

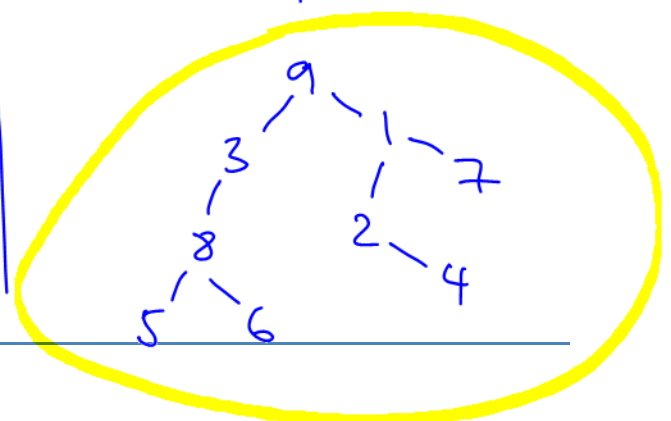
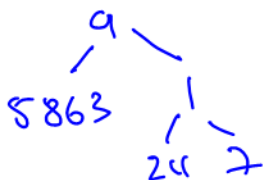
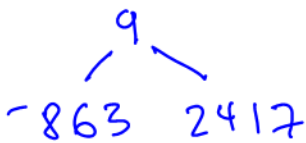
$$27 + x = |E| + 1$$

$$|E| = 26 + x$$

2. (1.5 marks) Given the following inorder and postorder of the tree T , recover (draw) the tree:

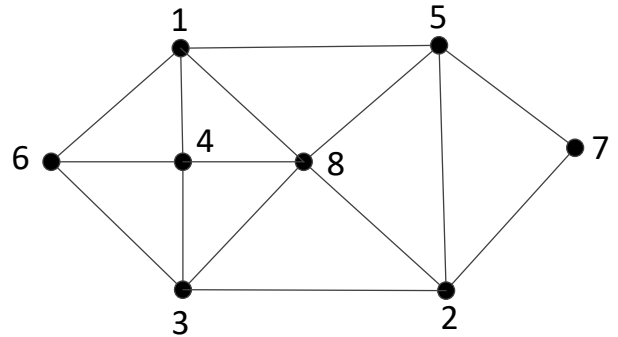
Inorder: 5, 8, 6, 3, 9, 2, 4, 1, 7

Postorder: 5, 6, 8, 3, 4, 2, 7, 1, 9



3. (2 marks) Do modified DFS on the following graph, **starting at vertex 1**:
For full marks, you must include both the correct Return Stack and the list of vertices printed on the screen.

```
dfs(vertex v)
{
    visit (v);
    for each neighbor w of v //in the increasing order
        if w is unvisited
        {
            print v
            dfs(w);
            print w
        }
}
```



Show the following:

1) RETURN STACK

[illegible]

2) PRINTED ON THE SCREEN:

[illegible]